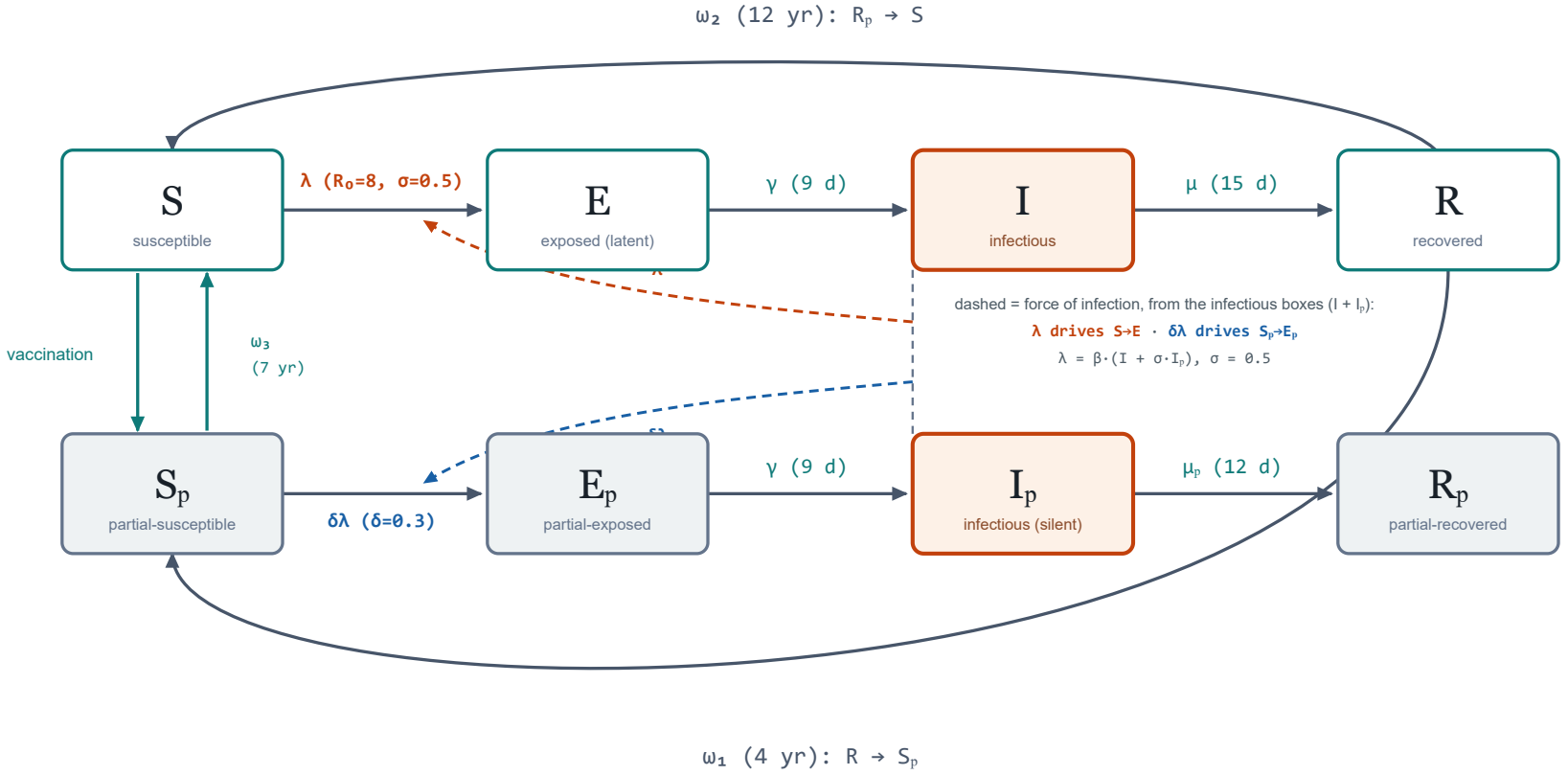
EPYSCENARIO · SOAR / IRS

Pertussis transmission model — SEIRS with partial-immunity track

8 compartments, 7 age bands (0-1, 1-6, 7-10, 11-19, 20-49, 50-64, 65+). Each transition is labelled with its symbol and **(default value)**. Structure adapted from Wearing & Rohani (2009).



SYMBOL	PARAMETER / TRANSITION	DEFAULT
R_0	Basic reproduction number (sets β on S→E)	**8**
γ	Incubation rate, E→I & E_p→I_p (= 1/incubation)	**9 days**
μ	Recovery rate, I→R (= 1/infectious)	**15 days**
μ_p	Recovery rate, I_p→R_p (partial, milder)	**12 days**
σ	Relative infectiousness of partial cases (I_p)	**0.5**
δ	Relative susceptibility of partial pool (S_p)	**0.3**

SYMBOL	PARAMETER / TRANSITION	DEFAULT
ω_1	Waning, R→S_p (recovered → partial)	**4 years**
ω_2	Waning, R_p→S (partial → susceptible)	**12 years**
ω_3	Vaccine waning, S_p→S	**7 years**
—	Seasonality peak day (day of year)	**240**
—	Seasonality amplitude	**Low**
S→S_p	Vaccination (moves susceptibles to partial pool)	**per schedule**

Initial conditions (defaults): infected 0.1% · immune 90% · partial-infection share 33% · partial-immune (S_p) share 30%. Force of infection on S is $\lambda = \beta \cdot (I + \sigma \cdot I_p)$; on S_p it is $\delta \cdot \lambda$. β is calibrated so the dominant next-generation eigenvalue equals R_0.

Conventions: solid arrows = flows of individuals (labelled with their per-capita rate); dashed arrows = the force of infection (an influence that depends on I and I_p, not a flow). **Closed population** — no births or deaths: susceptibility is replenished by waning ($\omega_1/\omega_2/\omega_3$), not by births, a simplification of the Wearing–Rohani framework (the engine represents compartment-to-compartment transitions only). **μ** and **μ_p** denote recovery rates here, not mortality.